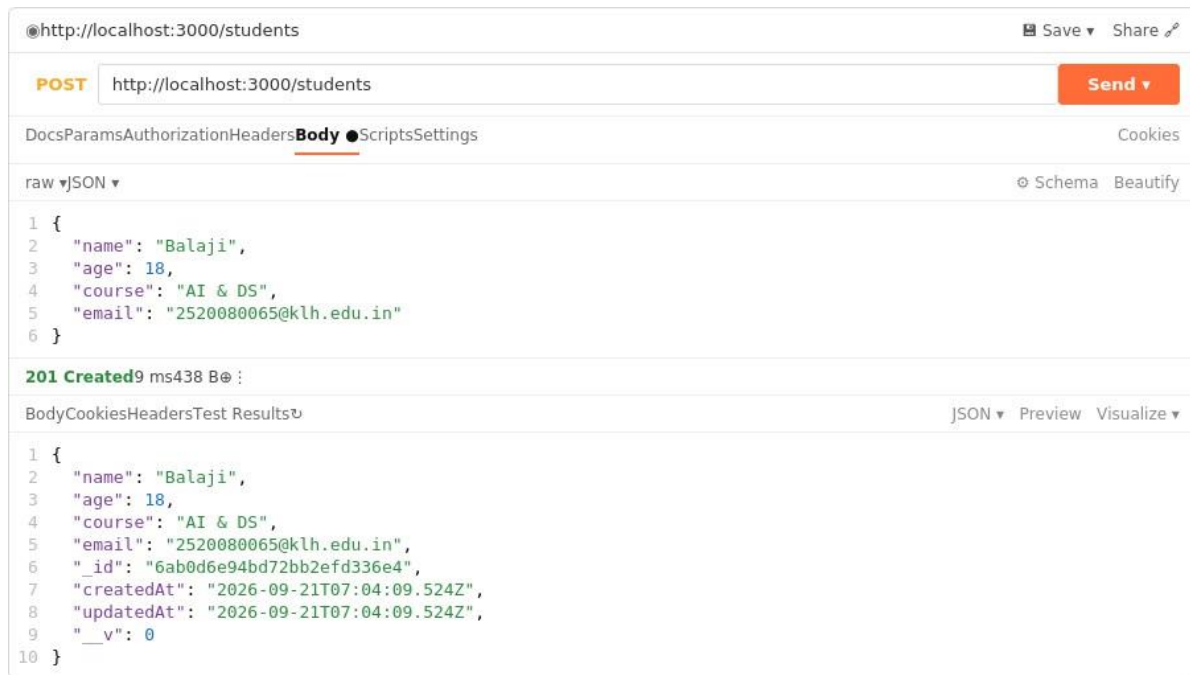


WEEK-7 SKILL — Student Records CRUD API [V Akhil, 2520080029]

POST:



http://localhost:3000/students

POST http://localhost:3000/students

Body

```
1 {
2   "name": "Balaji",
3   "age": 18,
4   "course": "AI & DS",
5   "email": "2520080065@klh.edu.in"
6 }
```

201 Created 9 ms 438 B

Body

```
1 {
2   "name": "Balaji",
3   "age": 18,
4   "course": "AI & DS",
5   "email": "2520080065@klh.edu.in",
6   "_id": "6ab0d6e94bd72bb2efd336e4",
7   "createdAt": "2026-09-21T07:04:09.524Z",
8   "updatedAt": "2026-09-21T07:04:09.524Z",
9   "__v": 0
10 }
```

After POST in mongosh:

```
{
  _id: ObjectId('6ab0d6e94bd72bb2efd336e4'),
  name: 'Balaji',
  age: 18,
  course: 'AI & DS',
  email: '2520080065@klh.edu.in',
  createdAt: ISODate('2026-09-21T07:04:09.524Z'),
  updatedAt: ISODate('2026-09-21T07:04:09.524Z'),
  __v: 0
}
```

GET all data:

http://localhost:3000/students Save Share

GET http://localhost:3000/students Send

Docs Params Authorization Headers **Body** Scripts Settings Cookies

raw JSON Schema Beautify

200 OK 8 ms 643 B

Body Cookies Headers Test Results JSON Preview Visualize

```
1 [
2   {
3     "name": "Balaji",
4     "age": 18,
5     "course": "AI & DS",
6     "email": "2520080065@klh.edu.in",
7     "_id": "6ab0d6e94bd72bb2efd336e4",
8     "createdAt": "2026-09-21T07:04:09.524Z",
9     "updatedAt": "2026-09-21T07:04:09.524Z",
10    "__v": 0
11  },
12  {
13    "name": "K Venkat Ram Reddy",
14    "age": 23,
15    "course": "MS in BA",
16    "email": "kvrreddy@gmail.com",
17    "_id": "6ab0d7b74bd72bb2efd336e5",
18    "createdAt": "2026-09-21T07:07:35.746Z",
19    "updatedAt": "2026-09-21T07:07:35.746Z",
20    "__v": 0
21  }
22 ]
```

GET all data in mongosh:

```
student_records> db.students.find()
[
  {
    _id: ObjectId('6ab0d6e94bd72bb2efd336e4'),
    name: 'Balaji',
    age: 18,
    course: 'AI & DS',
    email: '2520080065@klh.edu.in',
    createdAt: ISODate('2026-09-21T07:04:09.524Z'),
    updatedAt: ISODate('2026-09-21T07:04:09.524Z'),
    __v: 0
  },
  {
    _id: ObjectId('6ab0d7b74bd72bb2efd336e5'),
    name: 'K Venkat Ram Reddy',
    age: 23,
    course: 'MS in BA',
    email: 'kvrreddy@gmail.com',
    createdAt: ISODate('2026-09-21T07:07:35.746Z'),
    updatedAt: ISODate('2026-09-21T07:07:35.746Z'),
    __v: 0
  }
]
```

GET one with specific id:

http://localhost:3000/students/6ab0d6e94bd72bb2efd336e4 Save Share

GET http://localhost:3000/students/6ab0d6e94bd72bb2efd336e4 Send

DocsParamsAuthorizationHeadersBody ●ScriptsSettings Cookies

raw ▼JSON ▼ Schema Beautify

200 OK 9 ms433 B

BodyCookiesHeadersTest Results JSON Preview Visualize

```
1 {
2   "name": "Balaji",
3   "age": 18,
4   "course": "AI & DS",
5   "email": "2520080065@klh.edu.in",
6   "_id": "6ab0d6e94bd72bb2efd336e4",
7   "createdAt": "2026-09-21T07:04:09.524Z",
8   "updatedAt": "2026-09-21T07:04:09.524Z",
9   "__v": 0
10 }
```

PATCH → changed age from 23 to 25:

http://localhost:3000/students/6ab0ea72a7bcd067e137558c Save Share

PATCH http://localhost:3000/students/6ab0ea72a7bcd067e137558c Send

DocsParamsAuthorizationHeadersBody ●ScriptsSettings Cookies

raw ▼JSON ▼ Schema Beautify

200 OK 594 ms444 B

BodyCookiesHeadersTest Results JSON Preview Visualize

```
1 {
2   "age": 25
3 }

1 {
2   "name": "K Venkat Ram Reddy",
3   "age": 25,
4   "course": "MS in BA",
5   "email": "kvrreddy@gmail.com",
6   "_id": "6ab0ea72a7bcd067e137558c",
7   "createdAt": "2026-09-21T07:07:35.746Z",
8   "updatedAt": "2026-09-21T08:27:30.409Z",
9   "__v": 0
10 }
```

After modification of age output in mongosh:

```
{
  _id: ObjectId('6ab0ea72a7bcd067e137558c'),
  name: 'K Venkat Ram Reddy',
  age: 25,
  course: 'MS in BA',
  email: 'kvrreddy@gmail.com',
  createdAt: ISODate('2026-09-21T07:07:35.746Z'),
  updatedAt: ISODate('2026-09-21T08:27:30.409Z'),
  __v: 0
}
```

DELETE:



Observation of output in mongosh:

```

student_records> db.students.find()
[
  {
    _id: ObjectId('6ab0d6e94bd72bb2efd336e4'),
    name: 'Balaji',
    age: 18,
    course: 'AI & DS',
    email: '2520080065@klh.edu.in',
    createdAt: ISODate('2026-09-21T07:04:09.524Z'),
    updatedAt: ISODate('2026-09-21T07:04:09.524Z'),
    __v: 0
  }
]
student_records>

```

Folder Structure:

```

Week_7_StudentRecordsAPI
├── controllers
│   └── studentController.js
├── models
│   └── Student.js
├── routes
│   └── studentRoutes.js
├── server.js
└── package.json

```

Code:

package.json

```

{
  "name": "week_7_studentrecordsapi",
  "version": "1.0.0",
  "type": "module",
  "description": "",
  "main": "server.js",
  "scripts": {
    "start": "node server.js",
    "dev": "nodemon server.js"
  },
  "keywords": [],

```

```

"author": "",
"license": "ISC",
"dependencies": {
  "express": "^5.2.1",
  "mongoose": "^9.10.1"
},
"devDependencies": {
  "nodemon": "^3.1.14"
}
}

```

models/Student.js

```

import mongoose from 'mongoose';

const studentSchema = new mongoose.Schema({
  name: {
    type: String,
    required: true,
    trim: true
  },
  age: {
    type: Number,
    required: true,
    min: 16,
    max: 100
  },
  course: {
    type: String,
    required: true,
    trim: true
  },
  email: {
    type: String,
    required: true,
    unique: true,
    lowercase: true
  }
}, {
  timestamps: true
});

const Student = mongoose.model('Student', studentSchema);
export default Student;

```

controllers/studentController.js

```

import Student from '../models/Student.js';

export const createStudent = async (req, res) => {
  try {
    const student = await Student.create(req.body);
    res.status(201).json(student);
  } catch (error) {
    res.status(400).json({ error: error.message });
  }
};

export const getAllStudents = async (req, res) => {
  try {
    const students = await Student.find();
    res.status(200).json(students);
  } catch (error) {
    res.status(500).json({ error: error.message });
  }
};

```

```

export const getStudentById = async (req, res) => {
  try {
    const student = await Student.findById(req.params.id);
    if (!student)
      return res.status(404).json({ message: 'Student not found' });
    res.status(200).json(student);
  } catch (error) {
    res.status(400).json({ error: error.message });
  }
};

export const updateStudent = async (req, res) => {
  try {
    const student = await Student.findByIdAndUpdate(
      req.params.id,
      req.body,
      { new: true, runValidators: true }
    );
    if (!student)
      return res.status(404).json({ message: 'Student not found' });
    res.status(200).json(student);
  } catch (error) {
    res.status(400).json({ error: error.message });
  }
};

export const deleteStudent = async (req, res) => {
  try {
    const student = await Student.findByIdAndDelete(req.params.id);
    if (!student)
      return res.status(404).json({ message: 'Student not found' });
    res.status(200).json({
      message: 'Student deleted',
      student
    });
  } catch (error) {
    res.status(400).json({ error: error.message });
  }
};

```

routes/studentRoutes.js

```

import express from 'express';
import {
  createStudent,
  getAllStudents,
  getStudentById,
  updateStudent,
  deleteStudent
} from '../controllers/studentController.js';

const router = express.Router();

router.post('/', createStudent);
router.get('/', getAllStudents);
router.get('/:id', getStudentById);
router.patch('/:id', updateStudent);
router.delete('/:id', deleteStudent);

export default router;

```

server.js

```

import express from 'express';
import mongoose from 'mongoose';
import studentRoutes from './routes/studentRoutes.js';

```

```
const app = express();
const PORT = 3000;

app.use(express.json());

app.get('/', (req, res) => {
  res.json({
    message: 'Student Records API is running'
  });
});

app.use('/students', studentRoutes);

mongoose.connect('mongodb://127.0.0.1:27017/student_records')
  .then(() => {
    console.log('MongoDB connected');
    app.listen(PORT, () => {
      console.log(`Server running on http://localhost:${PORT}`);
    });
  })
  .catch((error) => {
    console.error('MongoDB connection failed:', error.message);
  });
```